

JEE MAIN REVISION MASTER: PERIODIC TABLE & PERIODICITY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 173 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Periodic Table And Periodicity - Core Application	173 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
-----------------	--

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Periodic Table And Periodicity - Core Application (Very High Yield — 173 PYQs)

Core Principle: Key problem pattern in Periodic Table And Periodicity.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Match List - I with List - II. List - I Electronic configuration of neutral atom (where $n=2$) List - II 1st Ionizatio..." [Ans: A]
- [2026 Evening Shift] Q#1 (MCQ): "The 1st ionization enthalpy for Mg is $+737\text{kJ/mol}$. The most probable estimated value of the 2nd ionization enthalp..." [Ans: C]
- [2026 Morning Shift] Q#1 (MCQ): "In period 4 of the periodic table, the elements with highest and lowest atomic radii are respectively...." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

Match List - I with List - II. List - I Electronic configuration of neutral atom (where $n=2$) List - II 1st Ionization Energy (kJ mol^{-1}) A. ns^2 I. 2080 B. ns^2np^1 II. 899 C. ns^2np^3 III. 800 D. ns^2np^6 IV. 1402 Choose the correct answer from the options given below :

- (A) -II, B-III, C-IV, D-I **[Correct]**
- (B) -IV, B-III, C-II, D-I
- (C) -III, B-II, C-IV, D-I
- (D) -III, B-II, C-I, D-IV

Correct Answer: A

Solution Analysis: Ionization energy is the energy needed to remove the most loosely held (outermost) electron from a neutral atom. Ionization energy generally increases when the effective nuclear charge increases. This happens because the nucleus attracts the valence electron more strongly, so more energy is required to remove it. Also, some electronic configurations are extra stable. Electrons in stable configurations are harder to remove, so their ionization energy is higher. Fully filled orbitals have greater ionisation energy due to stable electronic configuration. Order of I.E: $\text{fully-filled orbital}$